Year 10 Foundation

REVISION PACK

Graphs 1

Name: _____

Class: _____

Graphs 1 · Gradients and straight-line graphs (Lessons 3.1, 3.2, 3.3)

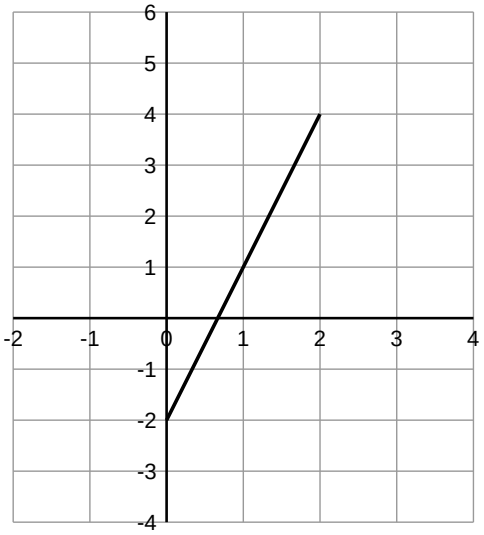
Worked example: gradient from a graph

A straight line is drawn on a grid. It passes through (0, 1) and (2, 7). Find the gradient of the line.

Step 1	Two points	$(0, 1)$ and $(2, 7)$
Step 2	Up	$7 - 1 = 6$
Step 3	Across	$2 - 0 = 2$
Step 4	Up ÷ across	$6 \div 2 = 3$

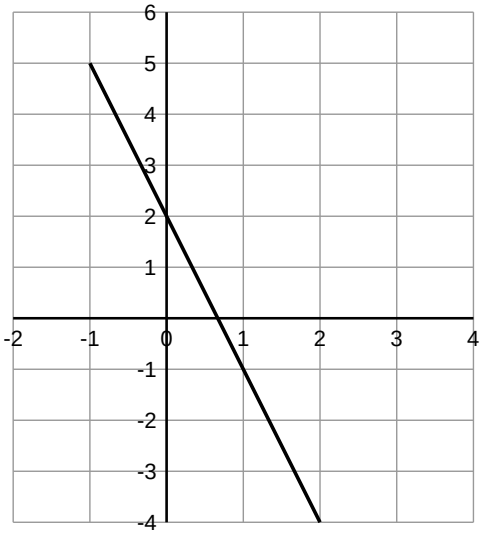
Practice: gradient from a graph

1.1



Find the gradient of the straight line drawn on the grid.

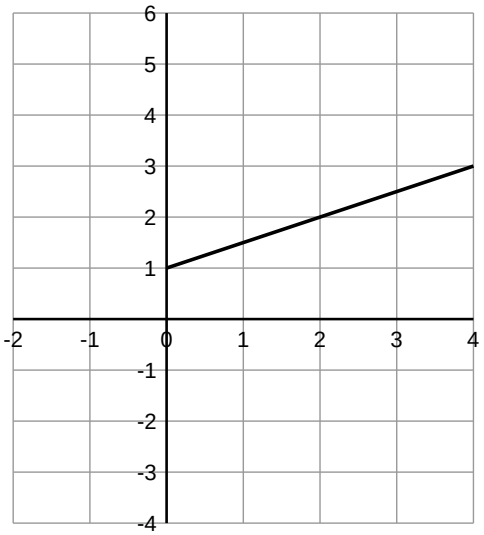
1.2



Find the gradient of the straight line drawn on the grid.

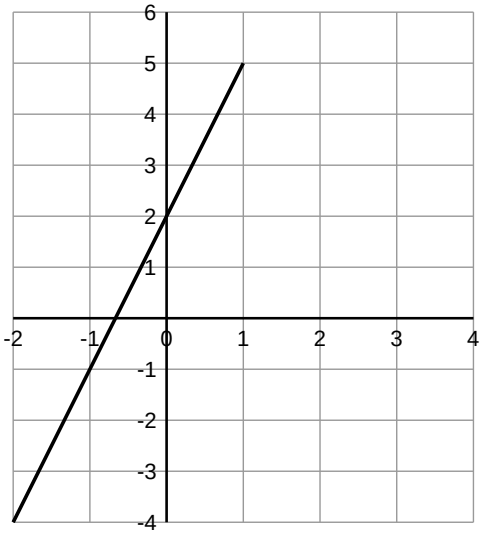
Practice: gradient from a graph (continued)

1.3



Find the gradient of the straight line drawn on the grid.

1.4



Find the gradient of the straight line drawn on the grid.

Graphs 1 · Gradients and straight-line graphs (Lessons 3.1, 3.2, 3.3)

Worked example: gradient from two points

A is the point (2, 1). B is the point (6, 9). Work out the gradient of the line AB.

Step 1	Change in y	$9 - 1 = 8$
Step 2	Change in x	$6 - 2 = 4$
Step 3	Gradient = up ÷ across	$8 \div 4 = 2$

Practice: gradient from two points

1.5 A is the point (1, 3). B is the point (4, 18). Work out the gradient of the line AB.

1.6 A is the point (2, 10). B is the point (5, 1). Work out the gradient of the line AB.

1.7 A is the point (0, 2). B is the point (8, 6). Work out the gradient of the line AB.

1.8 A is the point (−1, 4). B is the point (3, 16). Work out the gradient of the line AB.

1.9 A is the point (2, 7). B is the point (6, −1). Work out the gradient of the line AB.

1.10 A is the point (1, −2). B is the point (5, 10). Work out the gradient of the line AB.

Graphs 1 · Gradients and straight-line graphs (Lessons 3.1, 3.2, 3.3)

Worked example: drawing a straight-line graph

Here is a table of values for $y = 3x - 1$.

(a) Complete the table of values. (b) Draw the graph of $y = 3x - 1$ on a grid.

(a)	**Step 1**	Put $x = -1$ in	$y = 3 \times (-1) - 1 = -4$
	Step 2	Do each x	$-4,\ -1,\ 2,\ 5,\ 8$
(b)	**Step 1**	Plot the points	$(-1, -4),\ (0, -1),\ (1, 2) \ldots$
	Step 2	Join them	a straight line with a ruler

Practice: drawing a straight-line graph

1.11



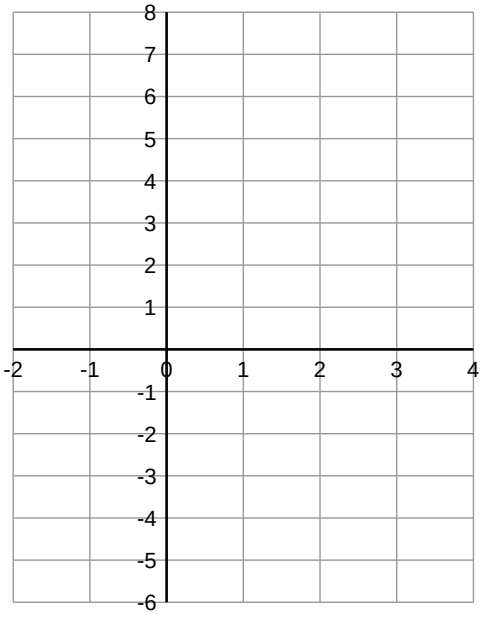
Here is a table of values for $y = 2x + 1$

x	−1	0	1	2	3
y		1		5	

(a) Complete the table of values.

(b) On the grid, draw the graph of $y = 2x + 1$ for values of x from −1 to 3

1.12



Here is a table of values for $y = 3x - 2$

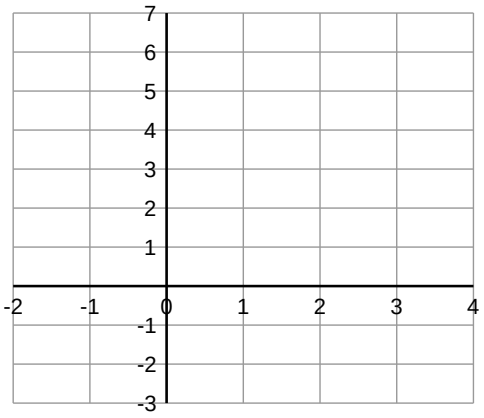
x	−1	0	1	2	3
y		−2		4	

(a) Complete the table of values.

(b) On the grid, draw the graph of $y = 3x - 2$ for values of x from −1 to 3

Practice: drawing a straight-line graph (continued)

1.13



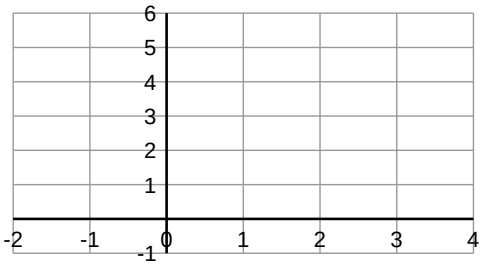
Here is a table of values for $y = -2x + 4$

x	−1	0	1	2	3
y		4		0	

(a) Complete the table of values.

(b) On the grid, draw the graph of $y = -2x + 4$ for values of x from −1 to 3

1.14



Here is a table of values for $y = x + 2$

x	−1	0	1	2	3
y		2		4	

(a) Complete the table of values.

(b) On the grid, draw the graph of $y = x + 2$ for values of x from −1 to 3

Graphs 1 · Gradients and straight-line graphs (Lessons 3.1, 3.2, 3.3)

Worked example: gradient and intercept

A straight line has equation $y = 5 - 2x$

(a) Write down the gradient of the line.

(b) Write down the coordinates of the point where the line crosses the y-axis.

(a)	**Step 1**	$y = mx + c$	$m = -2$
(b)	**Step 1**	c is where it crosses	$c = 5$
	Step 2	Answer	$(0, 5)$

Practice: gradient and intercept

1.15 A straight line has equation $y = 4x - 3$

(a) Write down the gradient of the line.

(b) Write down the coordinates of the point where the line crosses the y-axis.

1.16 A straight line has equation $y = 8 - 5x$

(a) Write down the gradient of the line.

(b) Write down the coordinates of the point where the line crosses the y-axis.

1.17 A straight line has equation $y = 6 - x$

(a) Write down the gradient of the line.

(b) Write down the coordinates of the point where the line crosses the y-axis.

1.18 A straight line has equation $y = 3x + 2$

(a) Write down the gradient of the line.

(b) Write down the coordinates of the point where the line crosses the y-axis.

1.19 A straight line has equation $y = 9 - 4x$

(a) Write down the gradient of the line.

(b) Write down the coordinates of the point where the line crosses the y-axis.

1.20 A straight line has equation $y = 6x - 1$

(a) Write down the gradient of the line.

(b) Write down the coordinates of the point where the line crosses the y-axis.

Graphs 1 · Parallel lines (Lesson 3.3)

Worked example: parallel lines

Write down the equation of a straight line that is parallel to $y = 3x - 2$

Step 1	Parallel lines have the same gradient	gradient $= 3$
Step 2	Choose a different y-intercept	e.g. 5
Step 3	Answer	$y = 3x + 5$

Practice: parallel lines

2.1 Write down the equation of a straight line that is parallel to $y = 2x + 1$

2.2 Write down the equation of a straight line that is parallel to $y = -4x + 3$

2.3 Write down the equation of a straight line that is parallel to $y = \frac{1}{2}x - 6$

2.4 Here are the equations of three straight lines.

$y = 5x + 1$ $y = 2x + 5$ $y = 5x - 3$

Which two lines are parallel? Give a reason for your answer.

2.5 Line L is parallel to $y = 3x + 7$ and passes through the point $(0, -1)$.

Write down the equation of line L.

2.6 Line M is parallel to $y = -2x + 5$ and passes through the point $(0, 4)$.

Write down the equation of line M.